

MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
Computer Vision-III [CSE_4031] SCHEME

PART-A

Answer all the questions.

- 1) A).A drone captures aerial images of a construction site to monitor progress and detect structural cracks. (5)

Illustrate how digital image processing steps help to extract structural details and detect possible defects.

- Image Acquisition
 - Drone captures the aerial image and imports it into the system.
- Image Enhancement
 - Contrast improvement, noise reduction, and clarity improvement.
- Image Restoration
 - Removal of blur, distortions and environmental noise.
- Color Image Processing
 - Use of color information to highlight structural surfaces and defects.
- Wavelet Processing
 - Multi-resolution representation for detecting fine crack details.
- Image Compression (optional)
 - Reducing storage or transmission size while retaining quality.
- Morphological Processing
 - Extraction and refinement of shapes; cleaning and enhancing crack structures.
- Image Segmentation
 - Dividing the image into meaningful regions; isolating crack regions.
- Feature Extraction
 - Detecting edges, boundaries, and measuring crack properties.
- Pattern Classification
 - Classifying the extracted features as “crack” or “non-crack”.
- Final Output
 - Highlighted structural cracks and defect detection results.

- B).Examine how Power-Law Transformations are used to control image contrast in photographs captured under varying illumination conditions.

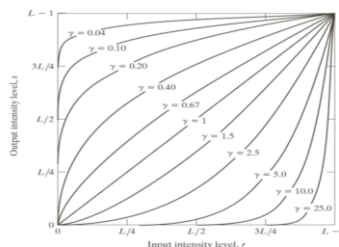
Power-law transformation

$$s = cr^\gamma$$

$\gamma = 1$ ideal case

$\gamma < 1$ maps narrow range of low intensity to wider range

$\gamma > 1$ opposite



- 2 A).Illustrate how edge detection techniques help in obtaining important structural details from digital images. (5)

Compare the working principles of the Laplacian and Canny edge detection methods

Edge detection helps to

- Identifies boundaries of objects and structural components
- Highlights discontinuities in intensity (cracks, edges of beams, joints)
- Extracts important structural shapes and outlines
- Supports further processes like segmentation and feature extraction
- Helps detect defects such as cracks, gaps, or misalignments

Comparison of Laplacian vs Canny Edge Detection

Laplacian Edge Detection

- Uses second-order derivative
- Detects regions of rapid intensity change
- Isotropic: responds equally in all directions
- Sensitive to noise (needs smoothing before use)
- Produces thick or noisy edges
- No direction information

Canny Edge Detection

- Uses gradient magnitude and direction (first derivative)
- Includes noise reduction using Gaussian smoothing
- Performs non-maximum suppression for thin edges
- Uses double thresholding to detect strong and weak edges
- Produces clean, thin, continuous edges
- More accurate and reliable than Laplacian

B).Apply the concept of the Histogram of Oriented Gradients descriptor to identify and represent the shape and structure of objects within an image. Discuss with a suitable real time application (5)

Gradient Computation

- Compute intensity gradients for each pixel
- Capture edge strength and direction
- Cell Formation
 - Divide image into small cells (e.g., 8×8 pixels)
 - Collect local gradient information
- Orientation Histogram
 - Gradients in each cell are grouped into orientation bins
 - Represents local shape and edge direction
- Block Normalization
 - Combine neighboring cells into blocks
 - Normalize histograms to reduce effects of brightness and shadows
- HOG Descriptor Formation
 - Concatenate all normalized histograms
 - Final feature vector represents object structure and contour

Real time applications-Pedestrian Detection in Autonomous Vehicles

Extract HOG features from road camera frames

Use classifier (e.g., SVM) to detect human shapes

Helps identify pedestrians based on contour and gradient patterns

Used for collision avoidance and road safety

3A).A drone captures rooftop images for inspection, but the photos appear misaligned and distorted due to the camera's tilt. Apply geometric transformations to adjust the tilted drone images and discuss in detail.(5)

Geometric transformations in image processing involve altering the spatial relationships between pixels in an image.

These transformations are used to manipulate the image's position, size, orientation, or shape.

The most common geometric transformations are:

Translation -explanation

Rotation-explanation

Scaling-explanation

Skewing (Shearing)-explanation

3B).In an AR game, players use their phone camera to find virtual coins, but lighting changes and camera shake cause feature mismatches, making the coins appear misplaced. Apply the RANSAC algorithm to detect and remove incorrect feature points, helping the game build a stable surface model and accurately place virtual coins on real surfaces. (5)

1. Randomly choose s samples. Typically s is the minimum samples to fit a model.
2. Fit the model to the randomly chosen samples.
3. Count the number M of data points (inliers) that fit the model within a measure of error ϵ .
4. Repeat Steps 1-3 N times
5. Choose the model that has the largest number M of inliers.

4A).Apply the learned motion estimation framework to detect abnormal motion patterns in video streams and discuss in detail. (5)

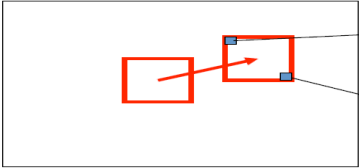
Video Input

- Continuous frames from surveillance/drone camera
 - Motion Estimation
 - Compute optical flow / motion vectors between frames
 - Feature Learning
 - Deep model (CNN/Autoencoder/LSTM) learns normal motion patterns
 - Motion Representation
 - Flow maps / trajectories describing speed & direction
 - Normal Pattern Modeling
 - Train system on normal behavior to form baseline motion profile
 - Anomaly Detection
 - Compare incoming motion with learned normal pattern
 - High deviation → abnormal motion
 - Output
 - Highlight abnormal regions (running, intrusion, panic, wrong-direction movement)

4)B. In a video tracking application, a system needs to estimate how an object moves between consecutive frames. Utilize the Lucas Kanade method to estimate the motion of objects across consecutive frames. (5)

$$I_x u + I_y v = -I_t \quad \longrightarrow \quad \begin{bmatrix} I_x & I_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = -I_t$$

- assume locally constant motion
 - ❖ pretend the pixel's neighbors have the same (u,v)
 - ✓ If we use a 5x5 window, that gives us 25 equations per pixel!



$$\begin{bmatrix} I_{x1} & I_{y1} \\ I_{x2} & I_{y2} \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} I_{t1} \\ I_{t2} \\ \vdots \end{bmatrix}$$

$$A\vec{u} = b$$

One of the most popular methods for computing optical flow is the Lucas-Kanade method. This method estimates the flow field by tracking the displacement of small patches of pixels within an image over time. The method starts by dividing the image into small patches, and then tracks the displacement of each patch between two consecutive frames. The displacement of each patch is determined by minimizing the difference between the patch in the first frame and the patch in the second frame.

5A).Develop an intruder detection system using face recognition for a secure facility where individuals may wear masks, glasses, or hats and explain the recognition pipeline to create a reliable metho that ensures accurate identification of unauthorized persons under varying appearance conditions. (5)

Input Acquisition

– CCTV/Surveillance camera captures face images (masked, glasses, hats).

• Preprocessing

– Face detection

– Alignment (eye/nose landmarks)

– Illumination & noise correction

• Occlusion Handling

– Mask/glasses/hat segmentation

– Extract only visible regions (upper face: eyes–eyebrows–forehead)

• Feature Extraction

– Deep embeddings using models like FaceNet / ArcFace

– Occlusion-robust features from periocular region

• Matching & Classification

– Compare extracted embeddings with authorized database

– Threshold-based similarity scoring

• Intruder Identification

– If match < threshold → flag as unauthorized

• Output / Alert

– Trigger alarm, notify security, store event log

5B).Apply a computer vision-based activity detection model to develop an elderly care monitoring system that observes indoor movements and detects signs of falls or distress. How would you apply these techniques to automatically identify normal vs. abnormal activities and trigger alerts in real time. (5)

• Video Capture – Indoor camera records continuous movement.

• Human Detection & Tracking – Locate and follow the person.

• Pose Estimation – Extract body keypoints (skeleton).

• Activity Analysis – Model learns normal vs. abnormal patterns.

• Fall/Distress Detection – Detect sudden collapse or unusual inactivity.

• Real-Time Alert – Send warning to caregivers when abnormal activity occurs.